Where K_{rot} represents the rotational kinetic energy expressed in terms of the body-fixed angular momentum vectors, K_{trans} is the translational kinetic energy, U is the potential energy which is a function of all the positions and dipole orientations. The vectors $\boldsymbol{\mu}_k$ can be taken to be unit vectors reflecting the dipole orientation, so they are subject to the constraint $\|\boldsymbol{\mu}_k\| \equiv 1$. The total dipole-dipole interaction potential then takes the form

$$\varphi_{ij}^{\text{dip}} = \varphi_{\text{LJ}}(r_{ij}) + \boldsymbol{\mu}_i^T \boldsymbol{L}(\boldsymbol{q}_{ij})\boldsymbol{\mu}_j,$$

with U given as the sum of these over pairs. The 3×3 matrix $\boldsymbol{L}$ is given by

$$\boldsymbol{L}(\boldsymbol{q}_{ij}) = \frac{1}{r_{ij}^3} - \frac{3}{r_{ij}^5}\boldsymbol{q}_{ij}\boldsymbol{q}_{ij}^T, \qquad \boldsymbol{q}_{ij} = \boldsymbol{q}_i - \boldsymbol{q}_j.$$

The rotational kinetic energy may be written

$$K_{\text{rot}}(\boldsymbol{\pi}_1, \boldsymbol{\pi}_2, \ldots, \boldsymbol{\pi}_{\text{N}}) = \frac{1}{2}\sum_{i=1}^{\text{N}} \frac{\boldsymbol{\pi}_{i,1}^2}{I_1} + \frac{\boldsymbol{\pi}_{i,2}^2}{I_2} + \frac{\boldsymbol{\pi}_{i,3}^2}{I_3},$$

with I_1, I_2, I_3 the moments of inertia, taken to be diagonal with respect to a suitable coordinate system. Introducing a rotation matrix $\boldsymbol{\Theta}_k$ for the kth dipole, we have $\boldsymbol{\mu}_k(t) = \boldsymbol{\Theta}_k\boldsymbol{\mu}_k(0)$, where $\boldsymbol{\mu}_k(0)$ is a reference configuration of the dipole (its initial state). Then

$$K_{\text{rot}} = \sum_{i=1}^{\text{N}} \text{Tr}\left(\dot{\boldsymbol{\Theta}}_i \boldsymbol{R} \dot{\boldsymbol{\Theta}}_i^T\right),$$

with $\boldsymbol{R}$ defined in terms of $\boldsymbol{T} = \text{diag}(I_1, I_2, I_3)$ by (4.40).

We now have two alternatives for representing the rotational motion. In the first, the rotation matrix describing the rigid body's orientation is the dynamical quantity (and we need to enforce the orthogonality of this matrix as a constraint). This leads to a constrained integration method, which makes use of SHAKE (in a somewhat different way from the method [84] which constrains the relative positions of particles). In the second formulation, the orthogonality is used to perform a reduction to an angular velocity (or momentum) representation with respect to the body frame of reference.

A Hamiltonian description of the motion may be formulated in either system of notation. The Hamiltonian formulation with rotation matrices involves variables $\boldsymbol{q}_{\text{cm}}, \boldsymbol{p}_{\text{cm}}$ (phase space variables describing the center of mass motion) and $\boldsymbol{\Theta}, \boldsymbol{\Pi}$, which are 3×3 matrices. The Hamiltonian is (for a single rigid body):

$$H_{\text{RB}} = \frac{\|\boldsymbol{p}_{\text{cm}}\|^2}{2M} + \frac{\text{Tr}\left(\boldsymbol{\Pi}\boldsymbol{R}^{-1}\boldsymbol{\Pi}^T\right)}{2} + U(\boldsymbol{q}_{\text{cm}}, \boldsymbol{\Theta}),$$

subject to the orthogonality constraint $\boldsymbol{\Theta}^T\boldsymbol{\Theta} = \boldsymbol{I}$. Here the potential energy has been expressed in terms of $\boldsymbol{q}_{\mathrm{cm}}$ and the orientation of the body. The constraint may be enforced using Lagrange multipliers with the following equations

$$\begin{aligned} \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{q}_{\mathrm{cm}} &= \boldsymbol{p}_{\mathrm{cm}}/M, \quad \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Theta} = \boldsymbol{\Pi}\boldsymbol{R}^{-1}, \\ \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{p}_{\mathrm{cm}} &= -\nabla_{\boldsymbol{q}_{\mathrm{cm}}} U, \; \tfrac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{\Pi} = -\nabla_{\boldsymbol{\Theta}} U - \boldsymbol{\Theta}\boldsymbol{\Lambda}, \\ & \boldsymbol{I} = \boldsymbol{\Theta}^T\boldsymbol{\Theta}. \end{aligned}$$

These equations may be solved using a SHAKE or RATTLE algorithm [84, 201]. However, we will present an alternative which is both conceptually simpler and easier to implement.

4.4.1 Euler Equations

An elegant form of the equations of motion for the free rigid body is expressed in terms of the angular momentum vector referred to the body frame.

The angular momentum vector in the reference frame is given by

$$\boldsymbol{l} = \sum_{i=1}^{\mathrm{N}} m_i \boldsymbol{\delta}_i \times \dot{\boldsymbol{\delta}}_i.$$

From Eq. (4.39) we may write this as

$$\boldsymbol{l} = \sum_{i=1}^{\mathrm{N}} m_i \boldsymbol{\delta}_i \times \boldsymbol{\omega} \times \boldsymbol{\delta}_i,$$

from which, using the vector triple product identity $\boldsymbol{a} \times \boldsymbol{b} \times \boldsymbol{c} = (\boldsymbol{a} \cdot \boldsymbol{c})\boldsymbol{b} - (\boldsymbol{a} \cdot \boldsymbol{b})\boldsymbol{c}$, we obtain

$$\boldsymbol{l} = \sum_{i=1}^{\mathrm{N}} m_i \left[\|\boldsymbol{\delta}_i\|^2 \boldsymbol{I} - \boldsymbol{\delta}_i \boldsymbol{\delta}_i^T \right] \boldsymbol{\omega}.$$

Define $\boldsymbol{\pi} = \boldsymbol{\Theta}^T\boldsymbol{l}$, the angular momentum vector referred to the body frame of reference, then we have (using $\boldsymbol{\delta}_i = \boldsymbol{\Theta}\delta_i(0)$):

$$\boldsymbol{\pi} = \boldsymbol{T}\boldsymbol{\omega}_{\mathrm{b}}.$$

Thus the inertia tensor relates the angular momentum vector in the body frame to the angular velocity vector in the body frame.